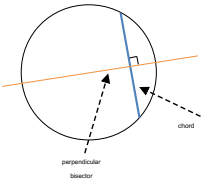
1.12 Coordinate Geometry: Circles

Chapter 6 · Original slide 4

Midpoints and Perpendicular Bisectors

4

Later in the chapter you will need to find the perpendicular bisector of a chord of a circle.



What two properties does a perpendicular bisector of two points $\mathbf{A}$ and $\mathbf{B}$ have?



Chapter 6 · Original slide 5

Test Your Understanding

5

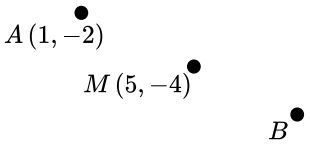
Find the perpendicular bisector of the line AB where A and B have the coordinates:
a) $A\,(4,7), B\,(10,17)$
b) $A\,(x_1,y_1), B\,(x_2,y_2)$

a

$$M\left(\tfrac{x_1+x_2}{2},\tfrac{y_1+y_2}{2}\right)$$
$$m_{AB}=\tfrac{y_2-y_1}{x_2-x_1} \qquad m_{\perp}=\tfrac{x_1-x_2}{y_2-y_1}$$
$$y-\tfrac{y_1+y_2}{2}=\tfrac{x_1-x_2}{y_2-y_1}\left(x-\tfrac{x_1+x_2}{2}\right)$$

b

A line segment AB is the diameter of a circle with centre $(5,-4)$. If A has coordinates $(1,-2)$, what are the coordinates of B?



We could use the formula $\left(\tfrac{x_1+x_2}{2},\tfrac{y_1+y_2}{2}\right)$, but it is easy to see from the diagram that whatever movement there is from A to M, we can continue to get to B.

1.12 Coordinate Geometry: Circles

Chapter 6 · Original slide 6

Exercise 6A - Y1 Page 115

6

6 The line segment JK is a diameter of a circle, where J and K are $(\frac{3}{4}, \frac{4}{3})$ and $(-\frac{1}{2}, 2)$ respectively. The centre of the circle lies on the line segment with equation $y = 8x + b$. Find the value of b.

7 The line segment AB is a diameter of a circle, where A and B are $(0, -2)$ and $(6, -5)$ respectively. Show that the centre of the circle lies on the line $x - 2y - 10 = 0$.

Chapter 6 · Original slide 7

Exercise 6B - Y1 Page 116

7

2 The line FG is a diameter of the circle centre C, where F and G are $(-2, 5)$ and $(2, 9)$ respectively. The line l passes through C and is perpendicular to FG. Find the equation of l. **(7 marks)**

3 The line JK is a diameter of the circle centre P, where J and K are $(0, -3)$ and $(4, -5)$ respectively. The line l passes through P and is perpendicular to JK. Find the equation of l. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

1.12 Coordinate Geometry: Circles

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Exercise 6B - Y1 Page 117

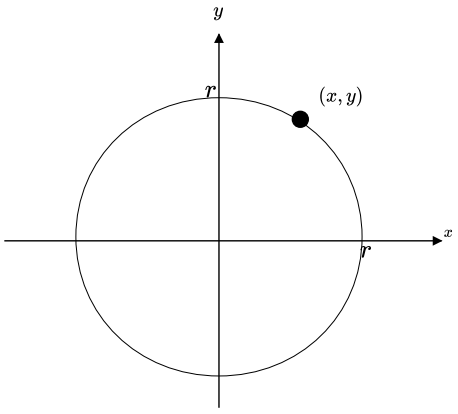
8

5 Point X has coordinates $(7, -2)$ and point Y has coordinates $(4, q)$. The perpendicular bisector of XY has equation $y = 4x + b$. Find the value of q and the value of b.

Chapter 6 · Original slide 9

Equation of a circle

9



Recall that a line can be a set of points (x, y) that satisfy some equation. Suppose we have a point (x, y) on a circle centred at the origin, with radius r. What equation must (x, y) satisfy?

(Hint: draw a right-angled triangle inside your circle, with one vertex at the origin and another at the circumference)

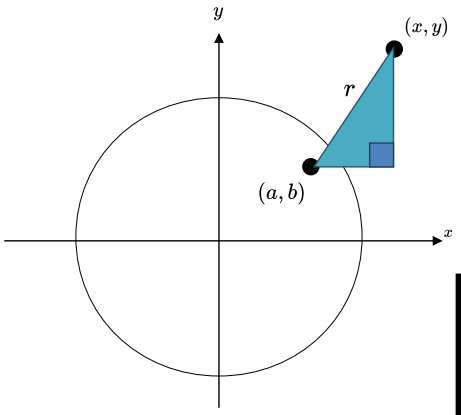
$$x^2 + y^2 = r^2$$

1.12 Coordinate Geometry: Circles

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Equation of a circle

10



Now suppose we shift the circle so it's now centred at (a, b). What's the equation now?

(Hint: What would the sides of this right-angled triangle be now?)

The equation of a circle with centre (a, b) and radius r is:

$$(x-a)^2 + (y-b)^2 = r^2$$

Chapter 6 · Original slide 11

Quickfire Questions

11

Centre	Radius	Equation
$(0, 0)$		

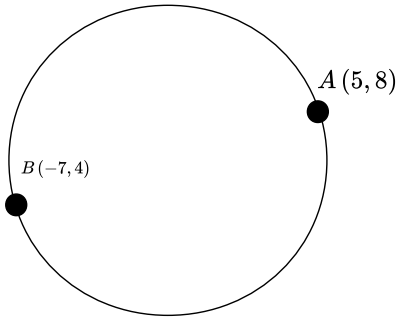
$(x+3)^2 + (y-5)^2 = 1$
$(x+5)^2 + (y-2)^2 = 49$
$(x+6)^2 + y^2 = 16$
$(x-1)^2 + (y+1)^2 = 3$
$(x+2)^2 + (y-3)^2 = 8$

1.12 Coordinate Geometry: Circles

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Finding the equation using points

12



A line segment AB is the diameter of a circle, where A and B have coordinates $(5, 8)$ and $(-7, 4)$ respectively. Determine the equation of the circle.

Hint: What two things do we need to use the circle formula?

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Test Your Understanding

13

Edexcel C2 Jan 2005 Q2

The points A and B have coordinates $(5, -1)$ and $(13, 11)$ respectively.
(a) Find the coordinates of the mid-point of AB. \hfill(2)
Given that AB is a diameter of the circle C,
(b) find an equation for C. \hfill(4)

1.12 Coordinate Geometry: Circles

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Completing the square

14

When the equation of a circle is in the form $(x-a)^2+(y-b)^2=r^2$, we can instantly read off the centre (a,b) and the radius r.
But what if the equation wasn't in this form?

Find the centre and radius of the circle with equation $x^2+y^2-6x+2y-6=0$

Hint: Have we seen a method in a previous chapter that allows us to turn a x^2 term and a x term into a single expression involving x?

Rearrange terms so that x terms are together and y terms are together.

Complete the square!

Chapter 6 · Original slide 15

Further Example

15

Edexcel C2 June 2012 Q3a,b

The circle C with centre T and radius r has equation
$x^2+y^2-20x-16y+139=0$

(a) Find the coordinates of the centre of C. \hfill(3)
(b) Show that $r=5$. \hfill(2)

1.12 Coordinate Geometry: Circles

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Example

16

Given the generic equation of a circle $x^2 + y^2 + ax + by + c = 0$ find a formula to determine the co-ordinate of the centre and the radius of the circle.

$$\left(x + \frac{a}{2}\right)^2 + \left(y + \frac{b}{2}\right)^2 = -c + \frac{a^2}{4} + \frac{b^2}{4}$$

$$r = \left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2 - c$$

$$c\left(-\frac{a}{2}, -\frac{b}{2}\right)$$

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Exercise 6C - Y1 Page 120

17

5 The line PQ is the diameter of the circle, where P and Q are $(5, 6)$ and $(-2, 2)$ respectively. Find the equation of the circle. \hfill(5 . . .

6 The point $(1, -3)$ lies on the circle $(x - 3)^2 + (y + 4)^2 = r^2$. Find the value of r.

1.12 Coordinate Geometry: Circles

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Exercise 6C - Y1 Page 120

18

8 a Show that $x^2 + y^2 - 4x - 11 = 0$ can be written in the form $(x - a)^2 + y^2 = r^2$, where a and r are numbers to be found. \hfill(2 marks)
b Hence write down the centre and radius of the circle with equation $x^2 +$
\hfill(2 marks)

9 a Show that $x^2 + y^2 - 10x + 4y - 20 = 0$ can be written in the form $(x - a)^2 + (y - b)^2 = r^2$, where a, b and r are numbers to be found. \hfill$(2\text{ m}\ldots$
b Hence write down the centre and radius of the circle with equation $x^2 + y^2 - 10x + 4y - 20 =$
$(2\text{ m}\ldots$

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Exercise 6C - Y1 Page 120

19

12 The point $P(7, -14)$ lies on the circle with equation $x^2 + y^2 + 6x - 14y = 17$. The point Q also lies on the circle such that PQ is a diameter. Find the coordinates of point Q.

13 The circle with equation $(x - k)^2 + y^2 = 41$ passes through the point $(3, 4)$.
Find the two possible values of k.

1.12 Coordinate Geometry: Circles

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Extension

20

1 [MAT 2009 1B] The point on the circle $x^2 + y^2 + 6x + 8y = 75$ which is closest to the origin, is at what distance from the origin?

2 [MAT 2007 1D] The point on the circle $(x-5)^2 + (y-4)^2 = 4$ which is closest to the circle $(x-1)^2 + (y-1)^2 = 1$ has what coordinates?

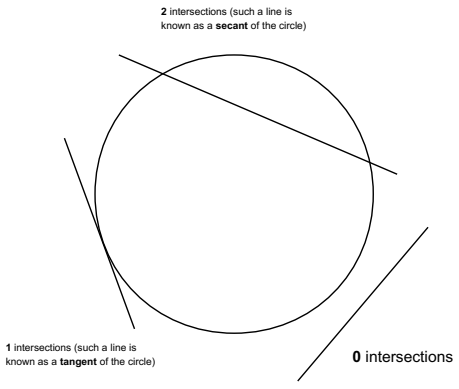
3 [MAT 2016 1I] Let a and b be positive real numbers. If $x^2 + y^2 \leq 1$ then the largest that $ax + by$ can equal is what? Give your expression in terms of a and b.

Chapter 6 · Original slide 21

Intersections of Lines and Circles

21

Recall that to consider the **intersection of two lines**, we attempt to solve them **simultaneously** by substitution, potentially using the **discriminant** to show that there are no solutions (and hence no points of intersection).



Show that the line $y = x + 3$ never intersects the circle with equation $x^2 + y^2 = 1$.

1.12 Coordinate Geometry: Circles

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Test Your Understanding

22

Find the points of intersection where the line $y = x + 6$ meets $x^2 + (y-3)^2 = 29$.

Using an algebraic (and not geometric) method, determine the k such that the line $y = x + k$ **touches** the circle with equation $x^2 + y^2 = 1$.

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Exercise 6D - Y1 Page 122

23

6 a Show that the line $x + y = 11$ meets the circle with equation $x^2 + (y-3)^2 = 32$ at only one point. \hfill$(4\ \mathrm{mar}\ldots$

b Find the coordinates of the point of intersection. \hfill$(1\ \mathrm{ma}\ldots$

1.12 Coordinate Geometry: Circles

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Exercise 6D - Y1 Page 122

24

7 The line $y = 2x - 2$ meets the circle $(x - 2)^2 + (y - 2)^2 = 20$ at A and B.

a Find the coordinates of A and B.

b Show that AB is a diameter of the circle.

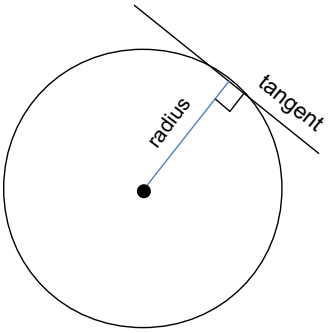
11 The line with equation $y = 4x - 1$ does not intersect the circle with equation $x^2 + 2x + y^2 = k$. Find the range of possible values of k.

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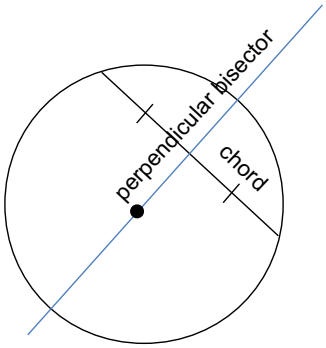
Tangents, Chords, Perpendicular Bisectors

25

There are two circle theorems that are of particular relevance to problems in this chapter, the latter you might be less familiar with:



The tangent is perpendicular to the radius (at the point of intersection).



The perpendicular bisector of any chord passes through the centre of the circle.

Why this will help:
If we knew the centre of the circle and the point of intersection, we can easily find the gradient of the radius, and thus the gradient and hence equation of the tangent.

Why this will help:
The first thing we did in this chapter is find the equation of the perpendicular bisector. If we had two chords, and hence found two bisectors, we could find the point of intersection, which would be the centre of the circle.

1.12 Coordinate Geometry: Circles

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Examples

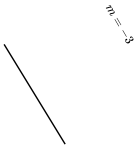
26

The circle C has equation
$(x-3)^2+(y-7)^2=100$.
a) Verify the point $P\,(11,1)$ lies on C.
b) Find an equation of the tangent to C at the point P, giving your answer in the form $ax+by+c=0$

Tip: Use 'subscripting' of variables to make clear to the examiner (and yourself!) what you're calculating.

A circle C has equation
$(x-4)^2+(y+4)^2=10$
The line l is a tangent to the circle and has gradient -3. Find two possible equations for l, giving your answers in the form $y=mx+c$.

This time we have the gradient, but don't have the points where the tangent(s) intersect the radius.



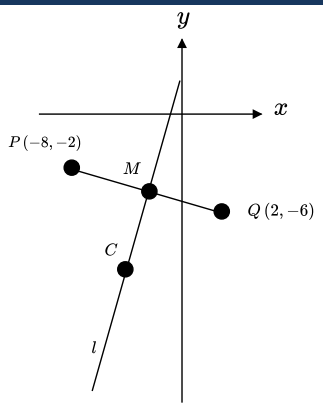
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Determining the Circle Centre

27

The points P and Q lie on a circle with centre C, as shown in the diagram. The point P has coordinates $(-8,-2)$ and the point Q has coordinates $(2,-6)$. M is the midpoint of the line segment PQ.
The line l passes through the points M and C.

a) Find an equation for l.
b) Given that the y-coordinate of C is -9:
i) show that the x-coordinate of C is -5.
ii) find an equation of the circle.



a

We know the line going through the midpoint of a chord and a centre of the circle is the perpendicular bisector of the chord.

b

c

1.12 Coordinate Geometry: Circles

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Test Your Understanding

28

A circle has centre $C\,(3,5)$, and goes through the point $P(6,9)$. Find the equation of the tangent of the circle at the point P, giving your equation in the form $ax+by+c=0$ where a, b, c are integers..

A circle passes through the points $A(0,0)$ and $B\,(4,2)$. The centre of the circle has x value -1. Determine the equation of the circle.

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Exercise 6E - Y1 Page 126-128

29

6 The tangent to the circle $(x+4)^2+(y-1)^2=242$ at $(7,-10)$ meets the y-axis at S and the x-axis at T.
a Find the coordinates of S and T. \hfill(5 marks)
b Hence, find the area of $\triangle OST$, where O is the origin. \hfill(3 marks)

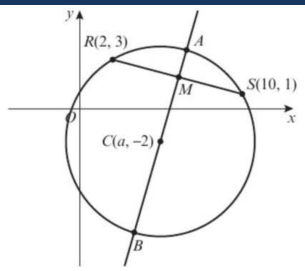
1.12 Coordinate Geometry: Circles

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Exercise 6E - Y1 Page 126-128

30

10 The points R and S lie on a circle with centre $C(a, -2)$, as shown in the diagram.
The point R has coordinates $(2, 3)$ and the point S has coordinates $(10, 1)$.
M is the midpoint of the line segment RS.
The line l passes through M and C.
a Find an equation for l. (4 marks)
b Find the value of a. (2 marks)
c Find the equation of the circle. (3 marks)



d Find the points of intersection, A and B, of the line l and the circle. (5 marks)

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Exercise 6E - Y1 Page 126-128

31

Extension:

1 *[MAT 2012 1A]* Which of the following lines is a tangent to the circle with equation $x^2 + y^2 = 4$?
A) $x + y = 2$
B) $y = x - 2\sqrt{2}$
C) $x = \sqrt{2}$
D) $y = \sqrt{2} - x$

2 *[AEA 2006 Q4]* The line with equation $y = mx$ is a tangent to the circle C_1 with equation $(x + 4)^2 + (y - 7)^2 = 13$
(a) Show that m satisfies the equation $3m^2 + 56m + 36 = 0$
The tangents from the origin O to C_1 touch C_1 at the points A and B.
(b) Find the coordinates of the points A and B.
Another circle C_2 has equation $x^2 + y^2 = 13$. The tangents from the point $(4, -7)$ to C_2 touch it at the points P and Q.
(c) Find the coordinates of either the point P or the point Q.
Mark scheme on next slide.

3 *[STEP 2005 Q6]*
(This is not a tangent/chord question but is worthwhile regardless!)
(i) The point A has coordinates $(5, 16)$ and the point B has coordinates $(4, -4)$. The variable P has coordinates (x, y) and moves on a path such that $AP = 2BP$. Show that the Cartesian equation of the path of P is $(x + 7)^2 + y^2 = 100$.
(ii) The point C has coordinates $(a, 0)$ and the point D has coordinates $(b, 0)$. The variable point Q moves on a path such that $QC = k \times QD$, where $k > 1$.
Given that the path of Q is the same as the path of P, show that $\frac{a+7}{b+7} = \frac{a^2+51}{b^2+51}$
Show further that $(a + 7)(b + 7) = 100$, in the case $a \neq b$.

Mark scheme on next slide.

1.12 Coordinate Geometry: Circles

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Mark Scheme for Extension Question 2

32

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Mark Scheme for Extension Question 3

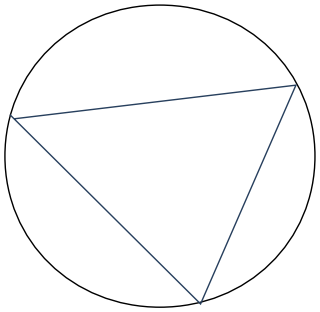
33

1.12 Coordinate Geometry: Circles

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Triangles in Circles

34



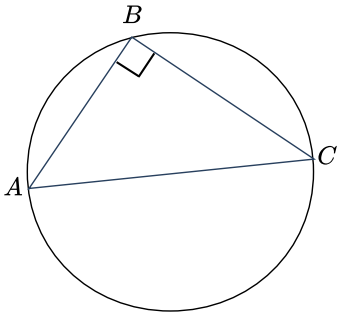
We'd say:

- The triangle **inscribes** the circle.
  (A shape inscribes another if it is inside and its boundaries touch but do not intersect the outer shape)
- The circle **circumscribes** the triangle.
- If the circumscribing shape is a circle, it is known as the **circumcircle** of the triangle.
- The centre of a circumcircle is known as the **circumcentre**.
-

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Triangles in Circles

35

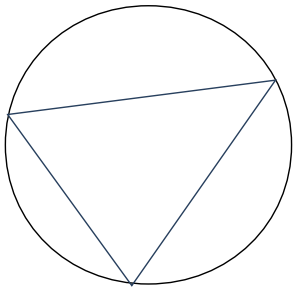


If $\angle ABC = 90^\circ$ then:

- $\mathbf{AC}$ **is the diameter of the circumcircle of triangle** $\mathbf{ABC}$**.**
-

Similarly if AC is the diameter of a circle:

- $\mathbf{\angle ABC = 90^\circ}$ **therefore** $\mathbf{AB}$ **is perpendicular to** $\mathbf{BC}$**.**
  - $\mathbf{AB^2 + BC^2 = AC^2}$



Given three points/a triangle we can find the centre of the circumcircle by:

- **Finding the equation of the perpendicular bisectors of two different sides.**
- **Find the point of intersection of the two bisectors.**

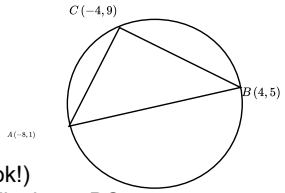
1.12 Coordinate Geometry: Circles

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Example

36

[Textbook] The points $A(-8,1), B(4,5), C(-4,9)$ lie on a circle.
a) Show that AB is a diameter of the circle.



Method 1:
Show that $AC^2 + BC^2 = AB^2$

Method 2: (not in textbook!)
Show that AC is perpendicular to BC.

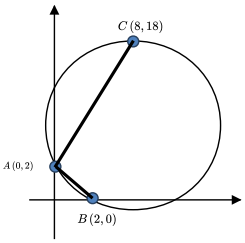
b) Hence find the equation of the circle.

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Example

37

The points $A(0,2), B(2,0), C(8,18)$ lie on the circumference of a circle. Determine the equation of the circle.



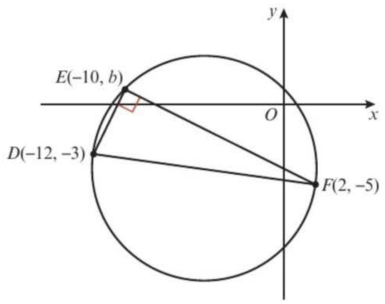
1.12 Coordinate Geometry: Circles

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Exercise 6F - Y1 Page 131-132

38

8 The points $D(-12, -3)$, $E(-10, b)$ and $F(2, -5)$ lie on the circle C as shown in the diagram.
Given that $\angle DEF = 90^\circ$ and $b > 0$
a show that $b = 1$ \hfill(5 marks)
b find an equation for C. \hfill(4 marks)



a $D(-12, -3)$, $E(-10, b)$, $F(2, -5)$

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Exercise 6F - Y1 Page 131-132

39

9 A circle has equation $x^2 + 2x + y^2 - 24y - 24 = 0$.
a Find the centre and radius of the circle.
b The points $A(-13, 17)$ and $B(11, 7)$ both lie on the circumference of the circle. Show that AB is a diameter of the circle.
c The point C lies on the negative x-axis and the angle $ACB = 90^\circ$. Find the coordinates of C.

1.12 Coordinate Geometry: Circles

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Exercise 6F - Y1 Page 131-132

40

Extension:

1 [STEP 2009 Q8 Edited] If equation of the circle C is $(x-2t)^2+(y-t)^2=t^2$, where t is a positive number, it can be shown that C touches the line $y=0$ as well as the line $3y=4x$.

Find the equation of the incircle of the triangle formed by the lines $y=0, 3y=4x$ and $4y+3x=15$.
Note: The incircle of a triangle is the circle, lying totally inside the triangle, that touches all three sides.